

Aram Khanlari . Jarvis Consulting

I am a data engineer with a strong academic background in algorithms, networks, and data processing, and I hold a Master of Computer Science from Concordia University. I have 4+ years of experience teaching computer science and mathematics, leading tutorials and labs on data structures, graph algorithms, databases, and network protocols. My interests include building reliable systems, analyzing data, and developing algorithmic solutions. As a software developer at Jarvis, I combine theory with hands-on implementation using Databricks, Python, SQL, Java, and Linux. I am passionate about solving real-world problems through analytical thinking, clean software practices, and clear communication.

Skills

Proficient: Python, Pandas/NumPy, Algorithm Design, RDBMS/SQL, Linux/Bash, Excel, Data Analysis

Competent: Java, Databricks, PySpark, Power BI, Agile/Scrum, Git, Docker, Machine Learning

Familiar: C++, C#, R, HTML/CSS, JavaScript

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_AramKhanlari

Python Data Analytics - London Gift Shop [GitHub]: Delivered a data analytics proof of concept for a UK-based online retailer to support business growth and decision-making. Extracted, transformed, and analyzed transactional data using Python to identify sales patterns and perform customer segmentation after sensitive data was sanitized and transferred to Jarvis servers. Produced insights, visualizations, and presented a 10-minute executive summary, followed by feedback and discussion with leadership and peers.

ETL pipeline in Databricks [GitHub]: Analyzed over 10 years of credit card transaction data to detect fraud patterns, behavioral anomalies, and long-term risk trends. Designed and implemented a scalable data pipeline using the Medallion architecture (Bronze, Silver, Gold layers) to ingest raw data, perform cleansing and validation, and produce analytics-ready datasets. Leveraged distributed processing frameworks to handle large-scale datasets efficiently and ensure reliability across environments. Automated data workflows to streamline ingestion, transformation, and scheduled refresh cycles. Integrated the pipeline with dynamic dashboards that automatically updated visualizations and key performance indicators. Generated and refined analytical questions to uncover actionable insights that informed fraud detection strategies and risk mitigation efforts.

Pyspark [GitHub]: Focused on Apache PySpark in distributed data processing environments. Rebuilt and scaled data analytics workflows that would normally run on a single machine, and made them suitable for larger datasets and production-like setups. Implemented using both Databricks and Zeppelin using Spark's structured APIs (DataFrames and SQL) to clean data, perform aggregations, and generate analytical insights efficiently.

Java Grep Application [GitHub]: Developed a Java application that mimics the Linux grep command to search matching strings within files, following a provided JavaGrep interface. Applied core Java concepts as part of the SDLC, including design, implementation, testing, and deployment. Performed manual testing, debugging via IDE tools, and optional JUnit tests. Packaged and distributed the application using the JVM, GitHub for source control, and Docker for deployment.

Linux Cluster Resource Monitoring App [GitHub]: Developed a Linux-based resource monitoring system on Rocky to collect CPU, memory, and disk usage metrics using Bash/Shell scripts and crontab. Containerized a PostgreSQL RDBMS using Docker to store host data and implemented analytical SQL queries for usage reporting and cluster health diagnostics. Followed a simplified SDLC to design, implement, test, and deploy the solution.

SQL [GitHub]: Designed relational schemas, created DDL tables, loaded test data, and executed analytical SQL queries using PostgreSQL. Practiced joins, aggregations, filtering logic, and string manipulation to reinforce RDBMS concepts and query optimization techniques.

Highlighted Projects

Broadcasting in Highly Connected Graphs: Developed theoretical bounds and practical algorithms for efficient message dissemination in highly connected networks. Implemented simulations using Python, Pandas, and NetworkX, validated algorithmic performance across graph classes, and created data visualizations. Published findings in the Discrete Applied Mathematics Journal ([here](#)) and completed as a Master's thesis ([here](#)).

Machine Learning Tools for Musical Harmonization: Designed and trained Logistic Regression and RNN-based models to generate harmonic accompaniment for melodies. Collected and cleaned musical datasets, implemented preprocessing pipelines, evaluated accuracy with statistical metrics, and iteratively improved model performance. Awarded ‘Best Paper’ among 40+ submissions.

Ant Colony Optimization for Routing: Applied Ant Colony Optimization to solve Traveling Salesman and shortest-path problems. Implemented network structures and distributed ants as agents. Defined pheromone update rules, heuristic scoring, and convergence criteria using Python, NumPy, and NetworkX. Produced performance plots and data visuals in Jupyter.

Professional Experiences

Software Developer, Jarvis (2025-Present): Collaborated in an Agile/Scrum environment to design and deliver backend and data-focused projects. Developed projects in Databricks and Apache Spark analysing data and generating visualizations. Implemented Linux/Shell automation, SQL analytics, Dockerized services, and version-controlled workflows. Gained hands-on experience with Linux/Shell, Git, Docker, and SDLC practices for real-world enterprise environments.

Online Mathematics & Computer Science Teacher, Dasa2.com (2022-Present): Taught OOP, data structures, network protocols, statistics, and calculus using Python, R, and Excel. Designed real-world programming assignments emphasizing data analysis, code efficiency, and problem-solving. Provided detailed feedback on debugging, algorithmic design, and best practices.

Piano Teacher & Curriculum Designer, Loisirs Ste. Dorothée (2024-Present): Led individual and group lessons for teenagers with varying skill levels, designing structured learning pathways and adapting instructional methods to student needs. Developed and coordinated a yearly piano curriculum across five sections and organized semester concerts. Strengthened communication, leadership, curriculum design, and stakeholder management skills directly transferable to team collaboration, technical mentorship, and project coordination.

Teaching & Lab Assistant (Computer Science), Concordia University (2021-2023): Delivered over 120 combined tutorials and lab sessions on algorithms, graph theory, data structures, AI heuristics, database systems, and network protocols. Guided more than 300 students by teaching algorithmic complexity, debugging code (Python, Java, SQL), and facilitating hands-on exercises in database management, TCP/UDP networking, multithreading, and synchronization. Designed assessments, created custom test cases, and provided structured feedback. Developed strong communication, technical explanation, and mentoring skills.

Education

Concordia University (2021-2023), Master of Computer Science, Gina Cody School of Engineering and Computer Science - NSERC and FRQ research grants - GPA: 4.15/4.3

American University of Armenia (2017-2021), Bachelor of Computer Science, Computer Science - Full academic scholarship - GPA: 3.85/4.0

Komitas State Conservatory (2017-2021), Bachelor of Music Composition, Music Composition and Theory - Full academic scholarship - GPA: 18.5/20

Miscellaneous

- Nomination for the Concordia Teaching Excellence Awards (2023)
- President, Graduate Research Activities Committee, Concordia University (2021-2023)
- Broadcasting in Highly Connected Graphs - Coauthored; published in Discrete Applied Mathematics (2025).
- Series-Parallel and Planar Graphs for Efficient Broadcasting - Coauthored; submitted to the International Journal of Foundations of Computer Science (under review).
- English (Fluent), French (Advanced), Armenian (Native), Russian (Advanced), Farsi (Basic)
- Piano performance and composition
- Board games, LEGO, Jigsaw puzzles